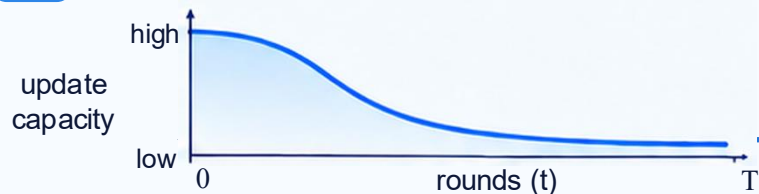


Proposal-side regularization

controls how search capacity is used

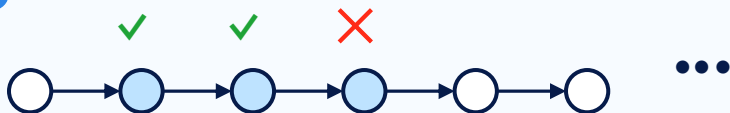
A Annealed update sparsity



Update budget decreases over time.

Early rounds allow broader edits; later rounds favor sparse, attributable changes.

B Evidence-aware credit assignment



Use the full history of gains and regressions, not just the latest win.

C Structured exploration



When progress stalls, redirect search toward underexplored components.

RRSI

Regularized transition rule

Proposal:

$$H' \sim P_{\text{reg}}(\cdot \mid H_t, \text{feedback})$$

Sample a regularized candidate H' from current harness and feedback history.

Selection:

$$H_{t+1} = \text{best admissible } H' \text{ or } H_t$$

Set H_{t+1} to the best admissible candidate if it improves robustly; otherwise keep H_t .

Selection-side regularization

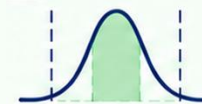
controls which gains may become permanent state

D Leakage screening



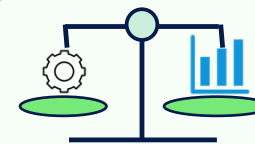
Reject benchmark-specific or task-specific logic.

E Noise-adjusted performance floor



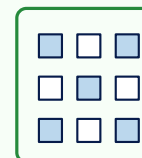
Prevent acceptance of gains that may be explained by evaluation noise.

F Complexity-aware acceptance (L_1 -style)



Extra complexity must be justified by measurable gain.

G Structural pruning (L_0 -style)



Prune components with no recent positive contribution.